Ve401 Midterm RC

Mathematica Part

Based on Zhang Fan's Mathematica Summary

Elements of Probability Theory

Permutation and Combination

- Permutation of k objects from A (total n elements)

In[ ]:=
```
Binomial[n,k]*k!
```

Out[ ]=
```
Binomial[7, 3] * 3!
```

210

- Combination of k objects from A (total n elements)

In[ ]:=
```
Binomial[n,k]
```

In[ ]:=
```
Binomial[7, 3]
```

Out[ ]=

35

PDF and CDF

- Basic Idea:

```
PDF[SomeDistribution[Parameters],value];
CDF[SomeDistribution[Parameters],value];
InverseCDF[SomeDistribution[Parameters],probability];
Mean[SomeDistribution[Parameters]];
Variance[SomeDistribution[Pariance]];
MomentGeneratingFunction[SomeDistribution[Parameters],variable/value];
```

- Example

In[ ]:=
```
PDF[NormalDistribution[μ, σ], x]
```

Out[ ]=

$$\frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi}\,\sigma}$$

In[ ]:=
```
CDF[NormalDistribution[μ, σ], x]
```
Out[ ]=

$$\frac{1}{2}\,\mathrm{Erfc}\left[\frac{-x+\mu}{\sqrt{2}\,\sigma}\right]$$

In[ ]:=
```
InverseCDF[NormalDistribution[μ, σ], 0.975]
```
Out[ ]=

$\mu + 1.95996\,\sigma$

In[ ]:=
```
Mean[NormalDistribution[μ, σ]]
```
Out[ ]=

μ

In[ ]:=
```
Variance[NormalDistribution[μ, σ]]
```
Out[ ]=

σ^2

In[ ]:=
```
MomentGeneratingFunction[NormalDistribution[μ, σ], t]
```
Out[ ]=

$e^{t\,\mu+\frac{t^2\,\sigma^2}{2}}$

For every distribution mentioned in the slide, Mathematica should have a corresponding Distribution. But be careful, distribution in Mathematica may be defined differently from the slide! (Especially when it comes to gamma function Γ)

Transformation of Variables

```
X=SomeDistribution[Parameters];
Y=SomeDistribution[Parameters];

Z=TransformedDistribution[Z(X,Y),{x≈X,y≈Y}];
Z
```

In[ ]:=
```
X = NormalDistribution[0, 1];
Y = NormalDistribution[0, 1];
(*Define the new variable Z=X^2+Y^2*)
Z = TransformedDistribution[x^2 + y^2, {x ≈ X, y ≈ Y}];
(*Display the distribution of Z*)
Z
```
Out[ ]=

ChiSquareDistribution[2]

Introduction to Statistics

In[10]:=
```
data := {79, 141, 228, 3, 20, 14, 97, 194, 28, 56, 75, 37, 122, 27, 10, 67, 23, 20, 103, 11,
   92, 99, 64, 6, 118, 136, 682, 4, 70, 11, 74, 40, 16, 114, 8, 149, 97, 7, 317, 346, 188,
   149, 68, 150, 88, 87, 155, 50, 26, 143, 126, 98, 153, 238, 30, 53, 132, 260, 296, 25, 61,
   87, 33, 51, 74, 111, 72, 178, 4, 67, 43, 229, 156, 117, 104, 27, 23, 23, 186, 524, 107,
   160, 41, 50, 352, 8, 153, 142, 306, 320, 85, 44, 116, 39, 264, 360, 192, 142, 44, 29};
```

Samples and Data - Basic Quantities

- Quartiles

In[ ]:=
```
Quartiles[data]
```

Out[ ]=

$$\left\{35, 87, \frac{299}{2}\right\}$$

- Interquartile Range

In[ ]:=
```
InterquartileRange[data]
```

Out[ ]=

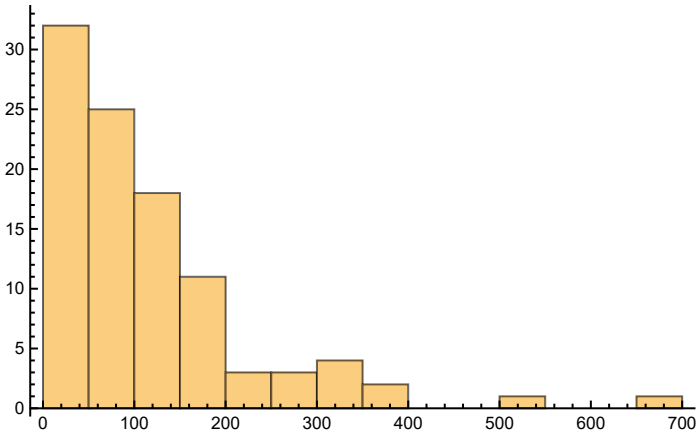
$$\frac{229}{2}$$

Histogram

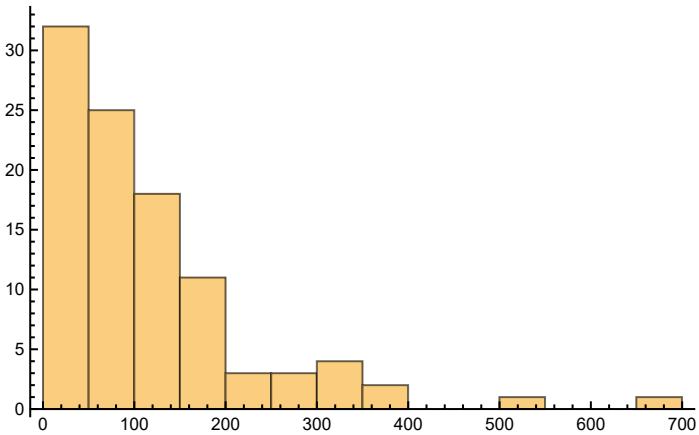
- Freedman-Diaconis Bin Width

In[4]:=
```
Histogram[data, "FreedmanDiaconis"]
Histogram[data, {"Raw", "FreedmanDiaconis"}]
```

Out[4]=



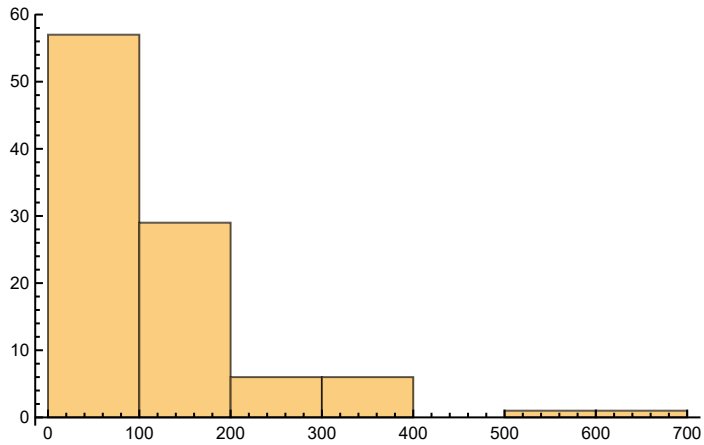
Out[5]=



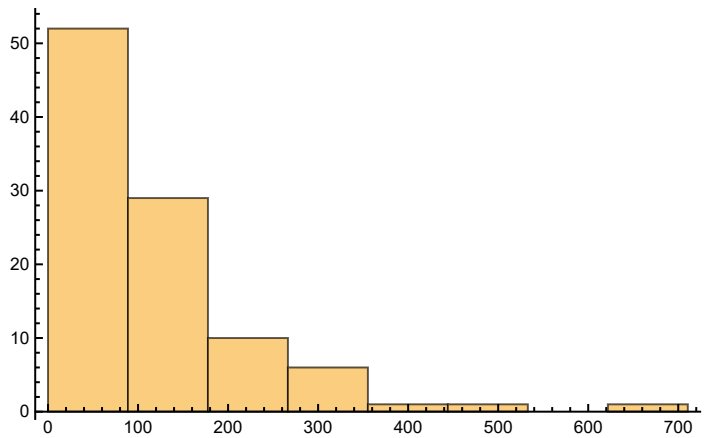
- Sturges's Rule Bin Number

In[6]:=
```
Histogram[data, "Sturges"]
Histogram[data, {"Raw", "Sturges"}]
```

Out[6]=



Out[7]=



Parameter Estimation

In[25]:=
```
FindDistributionParameters[data,
 ExponentialDistribution[β], ParameterEstimator → "MethodOfMoments"]
```

Out[25]=

{β → 0.0087382}

In[26]:=
```
FindDistributionParameters[data,
 ExponentialDistribution[β], ParameterEstimator → "MaximumLikelihood"]
```

Out[26]=

{β → 0.0087382}

Interval Estimation

In[27]:=
```
Data := {5.8, 6.0, 7.8, 7.4, 6.2, 6.1, 8.6, 9.7, 4.4, 4.4, 7.6, 9.0, 9.2, 6.4, 7.0, 8.7, 6.0,
   6.1, 6.8, 10., 5.3, 8.1, 8.9, 5.3, 6.7, 5.6, 3.7, 8.7, 6.6, 10.1, 6.9, 6.6, 6.4, 5.1};
```

- $z_{\alpha/2}$

```
InverseCDF[NormalDistribution[],1-α/2]
```

- $\chi^2_{\alpha/2}$

```
InverseCDF[ChiSquareDistribution[n],1-α/2]
```

- $t_{\alpha/2}$

```
InverseCDF[StudentTDistribution[n],1-α/2]
```

- Length, Sample Mean and Sample Variance of Data

```
Length[Data]
Mean[Data]
Variance[Data](*It's unbiased estimate. Not population variance!*)
```

Out[31]=

34

Out[32]=

6.97647

Out[33]=

2.77822

Example: compute the two-sided 95% confidence interval for the mean

Direct Computation:

In[74]:=

```
n = Length[Data];
α = .05;
mean = Mean[Data];
s = Sqrt[ Variance[Data]];
t = InverseCDF[StudentTDistribution[n - 1], 1 - α / 2];
L = t * s / √n;
mean - L
mean + L
```

Out[80]=

6.3949

Out[81]=

7.55804

MeanCI:

In[84]:=

```
Needs["HypothesisTesting`"]
MeanCI[Data, ConfidenceLevel → .95]
VarianceCI[Data, ConfidenceLevel → .90]
```

Out[85]=

{6.3949, 7.55804}

Out[86]=

{1.93421, 4.39369}